Homework 09

Question 1 Answers

We use 19 candidates: linear with $C \in \{0.01, 0.1, 1, 10\}$; polynomial of degree 2, 3, 4 with $C \in \{0.1, 1, 10\}$; and radial with $C \in \{0.1, 1, 10\}$, $\gamma \in \{0.01, 0.1\}$. Held fixed: polynomial kernel $(\gamma \mathbf{x}^\top \mathbf{x}' + r)^d$ with $\gamma = 1/9$ and $r = 1$, the 0.5 decision rule (sign of the decision value), and equal class weights.

```
library(e1071)

heart <- read.csv("data/saheart.csv")
heart$famhist <- factor(heart$famhist, levels = c("Absent", "Present"))
X <- data.matrix(heart[, names(heart) != "chd"])
y <- factor(heart$chd)
n <- nrow(X)
p <- ncol(X)

set.seed(432)
folds <- sample(rep(1:5, length.out = n))

cand <- rbind(
  data.frame(kernel = "linear", degree = NA, C = c(0.01, 0.1, 1, 10), gamma = NA),
  expand.grid(kernel = "polynomial", degree = 2:4, C = c(0.1, 1, 10), gamma = NA,
              stringsAsFactors = FALSE),
  expand.grid(kernel = "radial", degree = NA, C = c(0.1, 1, 10), gamma = c(0.01, 0.1),
              stringsAsFactors = FALSE)
)

fit_svm <- function(j, Xa, ya) {
  args <- list(x = Xa, y = ya, kernel = cand$kernel[j], cost = cand$C[j], scale = FALSE)
  if (cand$kernel[j] == "polynomial") {
    args <- c(args, degree = cand$degree[j], gamma = 1 / p, coef0 = 1)
  }
  if (cand$kernel[j] == "radial") args <- c(args, gamma = cand$gamma[j])
  do.call(svm, args)
}

fold_err <- sapply(1:5, function(k) {
  train <- folds != k
  mu <- colMeans(X[train, ])
  s <- apply(X[train, ], 2, sd)
  Xa <- scale(X[train, ], mu, s)
  Xb <- scale(X[!train, ], mu, s)
  sapply(seq_len(nrow(cand)), function(j) {
    mean(predict(fit_svm(j, Xa, y[train]), Xb) != y[!train])
  })
})

cand$cv_error <- rowMeans(fold_err)
cand$se <- apply(fold_err, 1, sd) / sqrt(5)

print(nrow(cand))
```

```
[1] 19
```

```
print(cbind(cand[, 1:4], round(cand[, 5:6], 3)), row.names = FALSE)
```

```
     kernel degree     C gamma cv_error    se
     linear     NA  0.01    NA    0.277 0.015
     linear     NA  0.10    NA    0.279 0.015
     linear     NA  1.00    NA    0.281 0.016
     linear     NA 10.00    NA    0.281 0.016
 polynomial      2  0.10    NA    0.271 0.013
 polynomial      3  0.10    NA    0.275 0.014
 polynomial      4  0.10    NA    0.281 0.011
 polynomial      2  1.00    NA    0.279 0.014
 polynomial      3  1.00    NA    0.318 0.016
 polynomial      4  1.00    NA    0.340 0.021
 polynomial      2 10.00    NA    0.294 0.017
 polynomial      3 10.00    NA    0.316 0.017
 polynomial      4 10.00    NA    0.359 0.010
     radial     NA  0.10  0.01    0.346 0.018
     radial     NA  1.00  0.01    0.262 0.015
     radial     NA 10.00  0.01    0.268 0.017
     radial     NA  0.10  0.10    0.344 0.020
     radial     NA  1.00  0.10    0.277 0.014
     radial     NA 10.00  0.10    0.340 0.019
```

```
lab <- with(cand, ifelse(kernel == "linear", "lin",
            ifelse(kernel == "polynomial", paste0("poly", degree),
                   paste0("rbf g=", gamma))))
lab <- paste0(lab, ", C=", cand$C)
cols <- c(linear = "#0072B2", polynomial = "#D55E00", radial = "#009E73")[cand$kernel]

par(mar = c(7, 4, 1, 1))
plot(seq_len(nrow(cand)), cand$cv_error, pch = 16, col = cols, xaxt = "n",
     ylim = range(cand$cv_error - cand$se, cand$cv_error + cand$se),
     xlab = "", ylab = "CV misclassification rate")
arrows(seq_len(nrow(cand)), cand$cv_error - cand$se, seq_len(nrow(cand)),
       cand$cv_error + cand$se, angle = 90, code = 3, length = 0.02, col = cols)
axis(1, at = seq_len(nrow(cand)), labels = lab, las = 2, cex.axis = 0.7)
abline(h = min(cand$cv_error), lty = 2)
```

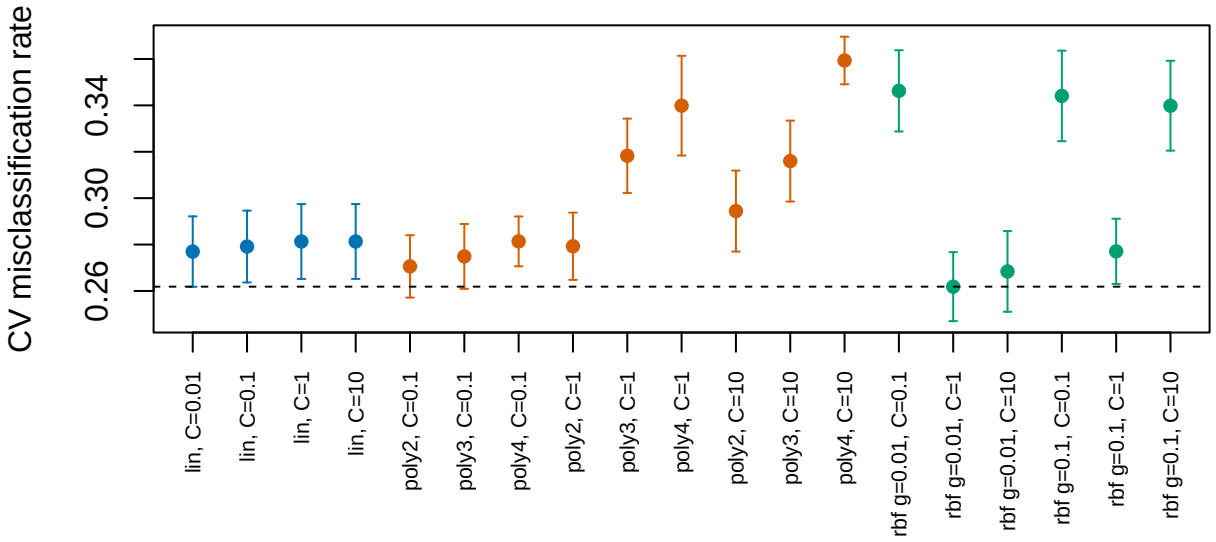


Figure 1: Mean 5-fold CV misclassification rate (±1 SE) for each candidate. Dashed line: smallest CV error.

```
best <- which.min(cand$cv_error)
print(cbind(cand[best, 1:4], round(cand[best, 5:6], 3)), row.names = FALSE)
```

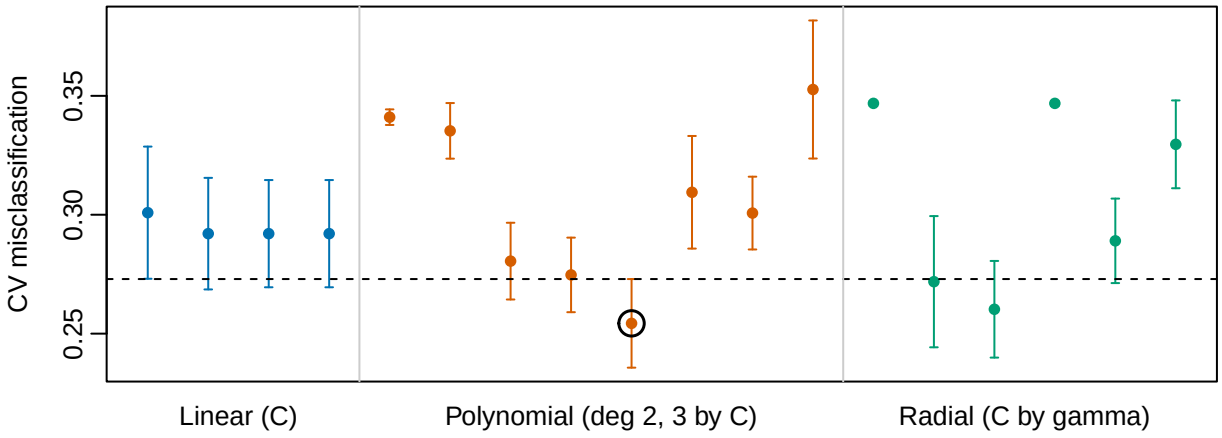
```
 kernel degree C gamma cv_error    se
 radial     NA 1  0.01    0.262 0.015
```

I prefer the **radial kernel with** $C = 1$**,** $\gamma = 0.01$, which has the smallest CV error (0.262, SE 0.015). Its small γ gives a smooth, nearly linear boundary, so it adds little complexity over the linear SVM (best 0.277 at $C = 0.01$, about one SE higher), which is a reasonable simpler alternative. Higher-degree polynomials and larger C overfit (errors up to 0.359), and radial with $C = 0.1$ (0.344–0.346) roughly matches always predicting `chd = 0` (160/462 = 0.346).

Question 2 Answers

The skill was invoked with: data `data/saheart.csv`, response `chd`, the nine other columns as predictors with `famhist` coded 1 = Absent, 2 = Present, seed 432, and output `svm-report/report.qmd`. The one-page, 11-point report follows. No peer review was received on my Homework 7 skill, so the new skill draws on that skill and the review I wrote in Homework 8: it fixes the page format, uses a one-SE rule with an explicit complexity order, and requires sensitivity and specificity.

Setup. We classify `chd` (0/1) from 9 predictors (`famhist` coded 1 = Absent, 2 = Present) in 462 cases with no missing values (302 / 160; minority class 35%, mild imbalance). With seed 432, a stratified 25% test set (116) is held out; 18 standardized SVMs (linear, polynomial degree 2–3, radial) are tuned by stratified 5-fold CV on the 346 training cases.



Model	CV error	SE	Chosen
linear, C = 0.1	0.292	0.023	
poly deg 2, C = 1	0.254	0.019	yes
radial, C = 10, gamma = 0.011	0.260	0.020	

Results. The lowest CV error is 0.254 (poly deg 2, C = 1); by the one-SE rule (dashed line, 0.273) we choose **poly deg 2, C = 1**, CV error 0.254 (SE 0.019). On the test set it misclassifies **0.276** (baseline 0.345), with sensitivity 0.450 and specificity 0.868; confusion matrix (rows observed 0/1): [66, 10; 22, 18].

Interpretation. The best nonlinear candidate (0.254) is more than one SE better than the best linear SVM (0.292), so a curved boundary helps. Large C with flexible kernels overfits. Test error is above CV error by 0.022, within the noise of 116 test cases. Sensitivity of 0.450 means 55% of `chd` cases are missed at the default cutoff.

Limitations. Results rest on one split and five folds, so the ranking of close candidates is unstable. The default decision rule favors the majority class; a cost-weighted rule would raise sensitivity at the expense of specificity.

SKILL.md

```
---
name: svm-report
description: >-
  Tune and compare SVM kernels for a classification dataset and write a one-page,
  11-point report. Use only when the user explicitly invokes svm-report and names
  the data, response, and predictors.
disable-model-invocation: true
---

# svm-report

Inputs from the user: data file, response, predictors and their coding, seed, output path.
If any is missing, ask; never guess variable roles. Use R (`e1071`) unless told otherwise.

## 1. Suitability
- Report n, p, class counts, missing values, and predictor types. Stop if the response
  is not categorical. Flag a minority class under 20%, p close to or above n, or missing
  data, and state how each affects the analysis.
- Apply the user's coding. SVMs are scale-sensitive, so always standardize predictors.

## 2. Kernels (one sentence each on why)
- Linear: the simplest baseline.
- Polynomial, degree 2 and 3 (`coef0 = 1`, `gamma = 1/p`): tests smooth interactions.
- Radial: tests a flexible, local boundary.

## 3. Tuning
- Linear and polynomial: cost C in {0.01, 0.1, 1, 10}.
  Radial: C in {0.1, 1, 10} x gamma in {0.1/p, 1/p}.
- Tune by stratified 5-fold CV on the training set, same folds for every candidate. In
  each fold, estimate standardization from the fitting part only; apply it to the rest.

## 4. Evaluation and selection
- Reserve a stratified 25% test set before any fitting; touch it only once, at the end.
- One-SE rule: among candidates within one SE of the lowest CV error, choose the simplest
  (linear < poly degree 2 < degree 3 < radial; then smaller C; then smaller gamma).
  Say if this changed the pick.
- Refit the chosen model on all training data. Report test error, confusion matrix
  (observed in rows), sensitivity, specificity, and the majority-class baseline error.

## 5. Figure and table
- Figure (at most 6.5 x 2.4 in): CV error +/- 1 SE for every candidate, grouped and
  colored by kernel (colorblind-safe), a dashed line at the minimum + 1 SE, and the
  chosen model circled.
- Table: best setting per kernel with CV error and SE (3 decimals); mark the chosen row.

## 6. Interpretation
- 3-4 sentences: whether a nonlinear kernel beat linear by more than one SE, how test
  error compares with CV error, and what sensitivity says about the minority class.
- If linear is chosen, name the 3 predictors with the largest |standardized weight|.

## 7. One-page report
- Quarto PDF, 11pt, 1 in margins, letter paper. Hide code (`echo: false`).
```

- Order: setup (2 sentences), figure, table, results, interpretation, limitations (2).
- Every number in the text is an inline code result. Render and confirm exactly one page; if longer, shorten text first, then the figure. Never drop the figure, table, or test.
- Return the seed, chosen model, CV and test errors, page count, and any issues.